

MICRO COMPUTER INSTITUTE

Web Design & Development

Foundation Study Pack - Volume 1

WEB | Duration: 6 Months

Learn - Practise - Apply

A structured bilingual-support handbook for classroom, lab and online learners.

Student use: Read each module, complete the lab activity, save evidence in your portfolio, and discuss doubts with your instructor.

Hindi support (Roman): Har module ko padhen, practical lab activity poori karen, apna work save karen aur doubt teacher se poochhen.

Course Roadmap

This 6 Months programme is organised into 8 learning modules. The sequence may be adjusted by the instructor to match the batch calendar.

No.	Module	Learning focus
1	Web Foundations	How websites work, browsers, servers, domains and hosting.
2	HTML	Semantic structure, text, links, images, tables and forms.
3	CSS	Selectors, box model, typography, colours and reusable styles.
4	Responsive Design	Flexbox, Grid, media queries and mobile-first layouts.
5	JavaScript Basics	Variables, functions, events and DOM interaction.
6	Backend Concepts	Requests, routing, validation, sessions and databases.
7	Publishing	Git basics, hosting, SSL, performance and SEO essentials.
8	Final Website	Design, build, test and deploy an accessible multi-page website.

Module 1: Web Foundations

How websites work, browsers, servers, domains and hosting.

Key ideas

- Explain the purpose and essential vocabulary of web foundations.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates web foundations. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 2: HTML

Semantic structure, text, links, images, tables and forms.

Key ideas

- Explain the purpose and essential vocabulary of html.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates html. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 3: CSS

Selectors, box model, typography, colours and reusable styles.

Key ideas

- Explain the purpose and essential vocabulary of css.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates css. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 4: Responsive Design

Flexbox, Grid, media queries and mobile-first layouts.

Key ideas

- Explain the purpose and essential vocabulary of responsive design.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates responsive design. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 5: JavaScript Basics

Variables, functions, events and DOM interaction.

Key ideas

- Explain the purpose and essential vocabulary of javascript basics.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates javascript basics. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 6: Backend Concepts

Requests, routing, validation, sessions and databases.

Key ideas

- Explain the purpose and essential vocabulary of backend concepts.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates backend concepts. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 7: Publishing

Git basics, hosting, SSL, performance and SEO essentials.

Key ideas

- Explain the purpose and essential vocabulary of publishing.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates publishing. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Module 8: Final Website

Design, build, test and deploy an accessible multi-page website.

Key ideas

- Explain the purpose and essential vocabulary of final website.
- Demonstrate the standard workflow accurately and safely.
- Choose suitable tools, settings and file formats for the task.
- Check the output, correct errors and save work with a meaningful name.

Lab activity

Create a small working example that demonstrates final website. Save the source file and a PDF or screenshot as evidence.

Self-check: Can you repeat the task without step-by-step help and explain why each important step is used?

Practical Workbook

Complete these tasks in order. The instructor may adapt names and sample data, but every learner should produce original work.

Practical 1. Create and save a correctly named practice file for Web Foundations. Include a short note describing what you learned.

Practical 2. Create and save a correctly named practice file for HTML. Include a short note describing what you learned.

Practical 3. Create and save a correctly named practice file for CSS. Include a short note describing what you learned.

Practical 4. Create and save a correctly named practice file for Responsive Design. Include a short note describing what you learned.

Practical 5. Create and save a correctly named practice file for JavaScript Basics. Include a short note describing what you learned.

Practical 6. Create and save a correctly named practice file for Backend Concepts. Include a short note describing what you learned.

Portfolio checklist

- Use folder format: CourseCode / StudentName / Module / Activity.
- Keep editable source files plus exported PDF or screenshot.
- Do not copy another student work; mention sources used.
- Back up important work in two locations.

Assignments & Assessment

Assignment 1: Concept Check

Write 300-500 words explaining three important concepts from Web Foundations and HTML. Add one real-life example.

Assignment 2: Practical Submission

Complete a combined task using CSS and Responsive Design. Submit source file, output and a one-page process note.

Assignment 3: Problem Solving

Identify a realistic institute or small-business problem that can be improved using Web Design & Development. Propose steps, tools, risks and success measures.

Assignment 4: Final Project

Build a complete project using at least four course modules. Test it, prepare screenshots, and present the result in 5-7 minutes.

Criterion	Weight
Accuracy and completeness	30%
Practical application	30%
Presentation and file organisation	20%
Originality, explanation and safe practice	20%

Revision & Safe Learning Guide

- Review class notes within 24 hours and practise again without watching the demonstration.
- Use legal software and trusted downloads. Scan external drives and downloaded files.
- Never share passwords, OTPs, private documents or student data.
- Verify AI-generated or internet information before using it in assignments.
- Use clear filenames, version numbers and regular backups.
- Ask for help when an error could damage data, equipment or accounts.

Quick viva questions

1. What is the purpose of Web Foundations, and what common mistake should a beginner avoid?
2. What is the purpose of HTML, and what common mistake should a beginner avoid?
3. What is the purpose of CSS, and what common mistake should a beginner avoid?
4. What is the purpose of Responsive Design, and what common mistake should a beginner avoid?
5. What is the purpose of JavaScript Basics, and what common mistake should a beginner avoid?
6. What is the purpose of Backend Concepts, and what common mistake should a beginner avoid?

Support

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This foundation pack supports instruction; software screens and rules may change. Follow current instructor demonstrations and official documentation.